Timestepping then proceeds as usual, with the slight modification that the time between steps will vary throughout integration. Numerical results with this method reported in [184] are very favorable.

3.7.3 Enhancements of the Collisional Verlet Method

A further idea introduced in [184] was to use a force decoupling strategy to reduce the cost of computations. The idea is to exploit the observation that impulsive forces ("kicks") can be supplied without reducing the order of accuracy as long as these have a vanishing component in the direction of the collision vector $\boldsymbol{u}_\perp$, that is if $\boldsymbol{F}_\perp = -\nabla U(\boldsymbol{q}_c) \cdot \boldsymbol{u}_\perp = 0$. This can be achieved in systems of spheres with pair potentials φ_{ij} only by writing $\varphi_{ij} = \alpha_{ij} + \beta_{ij}$ where the derivative of the second term is chosen to vanish at the point of contact between the spheres, i.e. $\alpha'_{ij}(\sigma_i + \sigma_j) = 0$. If this decomposition is used, then it is possible to build a 2nd order accurate hybrid method that uses only the first part α to define the quadratic Verlet paths for the collision detection scheme, whereas β is introduced as a standard "kick" at collision points.

In Collisional Verlet, the momenta are adjusted at the collision point to preserve the kinetic energy (and hence the total energy). It is possible to project to some other manifold using projection techniques like those mentioned previously, as discussed in [40]. In particular, one may use the backward error analysis to obtain a modified Hamiltonian $\tilde{H}_h$ corresponding to the Verlet method with stepsize h, then to project during collisions not onto the energy surface, but onto the modified energy surface, so that

$$\tilde{H}_h \equiv \text{const.}$$

(In practice, a low order approximation of $\tilde{H}_h$ is used, such as the truncation to terms of order four or six in the stepsize.) The resulting method substantially exhibits higher accuracy than the standard Collisional Verlet method in long term simulations, including accurate recovery of statistical properties (for example velocity autocorrelation functions). Additional cost is, however, involved in the handling of the collisions.

Other methods for hard sphere molecular dynamics and polymer modelling based on variational integration strategies, as well as comparative discussion, are provided in [235].

Exercises

1. Consider the solutions of